

qPCR Lab Workflow App

Final Mobile-First Copilot Design Document

Shared Lab Workspace for qPCR Experiment Management

1. Product Summary

This application is a mobile-first web application for research labs to manage qPCR experiments from planning to plate setup, raw Ct data upload, automated analysis, graph generation, notes, team sharing, and report export.

The application should not be built only as a qPCR calculator. It should be built as a **shared qPCR lab workflow workspace**.

Main interface:

```
Mobile phone first
Tablet second
Laptop/desktop third
```

The application should work well on mobile without requiring a laptop.

2. Product Positioning

Build this as:

```
A mobile-first shared qPCR experiment workspace for research labs, allowing
researchers to set up experiments, record observations, upload Ct data, run
calculations, generate graphs, export reports, and share work across the lab.
```

Do not build it as:

```
A desktop-only 96-well plate editor
```

or:

A personal qPCR calculator only

3. Main Problem

A qPCR experiment currently requires multiple disconnected steps:

1. Manual experiment setup.
2. Manual plate layout on paper or Excel.
3. Notes recorded separately.
4. Raw Ct values exported from qPCR instrument.
5. Manual copying into Excel.
6. Manual ΔCt , $\Delta\Delta Ct$, and fold-change calculations.
7. Manual graph preparation.
8. Manual report preparation.
9. Difficulty sharing experiment records with other lab members.

This causes:

- data entry errors,
- incorrect well-to-sample mapping,
- missing observations,
- inconsistent calculations,
- difficult reproduction of experiments,
- poor lab-level visibility,
- repeated work across researchers.

4. MVP Goal

The MVP should allow a researcher to:

1. Log in on mobile.
2. Access their lab workspace.
3. View a mobile lab dashboard.
4. Create or open a project.
5. Create a qPCR experiment.
6. Enter experiment metadata.
7. Set up a 96-well plate using mobile-friendly methods.
8. Upload raw Ct CSV data.
9. Map Ct values to wells.
10. Run qPCR analysis.
11. View results as mobile cards.
12. View simple fold-change graphs.

13. Add notes and observations.
 14. Export CSV, PNG, and PDF.
 15. Share visibility with lab members.
 16. Duplicate previous experiments.
-

5. Recommended Technology Stack

5.1 Preferred Stack

```
Frontend: Ionic React + TypeScript
Backend: ASP.NET Core 8 Web API
Database: Microsoft SQL Server
ORM: Entity Framework Core
Authentication: JWT
CSV Parsing: CsvHelper for .NET
Charts: Chart.js or Plotly.js
PDF Export: QuestPDF or HTML-to-PDF
File Storage: Local file system for MVP, cloud storage later
```

5.2 Why Ionic React

Since mobile phone is the main interface, Ionic React is recommended because it provides:

- mobile-style navigation,
- bottom tabs,
- mobile forms,
- touch-friendly controls,
- PWA support,
- later Android/iOS packaging option,
- React + TypeScript development model.

Alternative:

```
React + Material UI responsive design
```

This is acceptable if the goal is mobile-friendly web only, but Ionic is better if mobile is primary.

6. High-Level Architecture

```
Ionic React Mobile Frontend
    |
    | REST API
    v
ASP.NET Core & Web API
    |
    | Entity Framework Core
    v
MS SQL Server
    |
    | File metadata
    v
File Storage
    |
    | Uploaded CSV files
    | Exported PDF reports
    | Exported graph images
```

7. Core Data Hierarchy

Use this product hierarchy:

```
Lab
├── Users
├── Projects
│   └── Experiments
│       ├── Plate Layout
│       ├── Plate Wells
│       ├── Uploaded Ct Data
│       ├── Analysis Runs
│       ├── Analysis Results
│       ├── Graphs
│       ├── Reports
│       ├── Notes
│       └── Audit History
```

Every experiment must belong to a lab.

8. User Roles

8.1 System Role

SystemAdmin

Can:

- manage all labs,
- manage system users,
- view system-level logs.

This is not critical for the first MVP but should exist in the database.

8.2 Lab Roles

LabAdmin / PI

Can:

- view all lab experiments,
- manage lab users,
- create projects,
- edit lab experiments,
- finalize or unlock experiments,
- archive/delete experiments,
- export reports,
- view lab dashboard.

Researcher

Can:

- create experiments,
- edit own experiments,
- view lab-shared experiments,
- duplicate experiments,
- upload CSV files,
- run analysis,
- add notes,
- export reports.

Viewer

Can:

- view allowed experiments,
- view results,
- export if allowed,
- cannot edit experiments.

9. Sharing Model

Each experiment should have visibility:

Private
Project
Lab

Recommended MVP Default

Default Visibility = Lab

This supports lab sharing from the beginning.

View Rules

User can view an experiment if:

- user is SystemAdmin, or
- user is LabAdmin/PI of the lab, or
- experiment visibility is Lab and user belongs to lab, or
- experiment visibility is Project and user belongs to project, or
- user is experiment owner.

Edit Rules

User can edit an experiment if:

- experiment is not finalized,
- and user is owner, PI, LabAdmin, or project lead.

10. Experiment Status

Use these statuses:

Draft
PlateDesigned
DataUploaded
Analyzed
Finalized
Archived

Finalized Status

When experiment is finalized:

- plate layout cannot be edited,
- Ct data cannot be changed,
- uploaded CSV cannot be replaced,
- analysis result remains frozen,
- reports remain traceable,
- user can duplicate experiment to start a new version.

Only PI/LabAdmin should unlock finalized experiments.

11. Mobile-First Design Principle

The app must be usable without a laptop.

Important rule:

Do not make full 96-well grid editing the primary workflow on mobile.

The 96-well grid can exist, but mainly for:

- visual review,
- checking status,
- tapping one well,
- seeing warnings,
- confirming layout.

Primary mobile data entry should use:

1. Templates.
2. Quick Fill by Range.
3. Upload Plate Layout CSV.
4. Copy Previous Experiment.
5. Manual single-well correction.
6. Plate List View.

12. Mobile Navigation

Use bottom navigation:

Home
Projects
Experiments
Upload
More

Inside an experiment, use step-based workflow:

Details → Plate Setup → Upload Data → Analysis → Graphs → Notes → History

Avoid desktop-style top tab overload.

13. Main Mobile Screens

13.1 Login Screen

Fields:

Email
Password

Actions:

Login

13.2 Mobile Lab Dashboard

Use cards, not wide tables.

Example:

Lab Dashboard

[+ New Experiment]

Summary Cards:

- My Drafts: 3
- Pending Analysis: 5
- Experiments With Warnings: 2
- Recent Reports: 4

Recent Experiments:

CCR2_Repeat_1

Project: CCR2 Study

Owner: Kishi

Status: Analyzed

Warnings: 1

[Open]

Pilot_Run_03

Project: Drug Test

Owner: Maria

Status: Data Uploaded

[Open]

Dashboard should include:

- my drafts,
- pending analysis,
- experiments with warnings,
- recent lab experiments,
- recently exported reports.

13.3 My Experiments Screen

Card layout:

My Experiments

CCR2_Repeat_1

Project: CCR2 Study

Status: Analyzed

Last Updated: Today

[Open] [Duplicate]

Pilot_Run

Project: Pilot

Status: Draft

[Open]

13.4 Projects Screen

Project card:

CCR2 Inflammation Study

Experiments: 12

Researchers: 3

Latest: CCR2_Repeat_1

[Open Project]

Project detail should show:

- project objective,
- members,
- experiments,
- latest results,
- recent reports.

13.5 Create Experiment Screen

Fields:

Experiment Name

Project

Experiment Date

Objective

Hypothesis

Sample Source

Treatment Condition
Instrument Name
Reference Gene
Control Sample Name
Visibility
Notes

Defaults:

Plate Type = 96-well
Visibility = Lab
Status = Draft
Owner = logged-in user
Lab = current lab

When experiment is created:

- create experiment,
- create one 96-well plate layout,
- create 96 empty plate wells.

14. Mobile Plate Setup

14.1 Plate Setup Method Selection

When user opens Plate Setup, show:

How do you want to set up the plate?

1. Use Template
2. Quick Fill by Range
3. Upload Plate Layout CSV
4. Copy Previous Experiment
5. Manual Well Entry
6. Review Plate

This is the most important mobile-first correction.

14.2 Method 1: Use Template

Flow:

```
Select Template
Preview Plate
Apply to Experiment
Edit if needed
```

Example templates:

```
CCR2 Standard Layout
Treatment vs Control Layout
3 Replicate Layout
NTC Control Layout
```

Templates can be Phase 2, but the design should support this from the beginning.

14.3 Method 2: Quick Fill by Range

This should be included in MVP.

Example screen:

```
Quick Fill

From Well: A01
To Well: A03

Sample Name: S1
Target Gene: CCR2
Reference Gene: GAPDH
Sample Type: Test
Replicate Group: S1_CCR2

[Apply]
```

Example usage:

```
A01-A03 = S1 CCR2
A04-A06 = S2 CCR2
```

```
B01-B03 = S1 GAPDH
B04-B06 = S2 GAPDH
H10-H12 = NTC CCR2
```

This is much easier on mobile than selecting tiny grid cells.

14.4 Method 3: Upload Plate Layout CSV

Support mobile upload of layout CSV.

Expected layout CSV columns:

```
Well
SampleName
TargetGene
ReferenceGene
SampleType
ReplicateGroup
Notes
```

Example:

```
Well,SampleName,TargetGene,ReferenceGene,SampleType,ReplicateGroup
A01,S1,CCR2,GAPDH,Test,S1_CCR2
A02,S1,CCR2,GAPDH,Test,S1_CCR2
A03,S1,CCR2,GAPDH,Test,S1_CCR2
B01,S1,GAPDH,GAPDH,Reference,S1_GAPDH
```

14.5 Method 4: Copy Previous Experiment

Flow:

```
Select previous experiment
Preview layout
Copy layout only
Apply to current experiment
```

Copy:

Sample names
Target genes
Reference genes
Sample types
Replicate groups
Plate layout

Do not copy:

Ct values
Uploaded files
Analysis results
Warnings
Reports

14.6 Method 5: Manual Well Entry

For corrections only.

Screen:

Well: A01
Sample: S1
Target: CCR2
Reference: GAPDH
Type: Test
Replicate: S1_CCR2
Notes: Optional

[Save]
[Previous Well]
[Next Well]
[Copy From Previous]
[Clear Well]
[Exclude Well]

14.7 Plate Review Screen

Mobile plate review should support:

Zoomable grid
Horizontal scroll
Tap well to view details
Color-coded wells
Show only warnings
Filter by sample
Filter by target
Filter by missing Ct

The grid is for review, not primary entry.

14.8 Plate List View

Mobile list view is required.

Example:

Plate Wells

A01	S1	CCR2	Test	Ct: 23.45
A02	S1	CCR2	Test	Ct: 23.60
A03	S1	CCR2	Test	Ct: 23.52
B01	S1	GAPDH	Reference	Ct: 20.10
B02	S1	GAPDH	Reference	Ct: 20.20

Filters:

Sample
Target
Type
Missing Ct
Warnings
Excluded

15. Upload Ct Data on Mobile

The user should be able to upload a qPCR CSV file from mobile.

Screen:

Upload Ct CSV

[Upload CSV File]

Preview:

A01 - 23.45 - Valid

A02 - 23.60 - Valid

H12 - Undetermined - Warning

Column Mapping:

Well Column: [Well]

Ct Column: [Ct]

Sample Column: [Optional]

Target Column: [Optional]

[Map to Plate]

Use card/list preview, not wide table.

16. qPCR Analysis Logic

16.1 Core Formulas

$\Delta Ct = Ct(\text{target gene}) - Ct(\text{reference gene})$

$\Delta\Delta Ct = \Delta Ct(\text{sample}) - \Delta Ct(\text{control sample})$

$\text{Fold Change} = 2^{(-\Delta\Delta Ct)}$

$\text{Log2 Fold Change} = -\Delta\Delta Ct$

16.2 Calculation Rules

1. Excluded wells are ignored.
 2. Blank wells are ignored.
 3. NTC wells are not used in fold-change calculation.
 4. Reference gene is selected at experiment level.
 5. Control sample is selected at experiment level.
 6. Technical replicates are averaged.
 7. Standard deviation is calculated for replicate groups.
 8. Every analysis run is saved separately.
 9. Previous analysis runs are never overwritten.
-

17. Mobile Analysis Screen

Use summary cards instead of wide tables.

Example:

Analysis Complete

Experiment: CCR2_Repeat_1

Target: CCR2

Reference: GAPDH

Control: Control

S1

Δ Ct: 2.6

$\Delta\Delta$ Ct: 0.3

Fold Change: 0.81

S2

Δ Ct: 3.3

$\Delta\Delta$ Ct: 1.0

Fold Change: 0.50

Warning: High Ct

Add:

[View Full Table]

[View Warnings]

[View Graph]

[Export]

18. Warning Rules

Generate warnings for:

Missing Ct value

Invalid Ct value

High Ct value

High replicate variation

NTC amplification

Missing control sample

Missing reference gene
Missing target gene
No matching reference Ct for sample
No matching control ΔCt
Duplicate well in CSV
CSV well not found in plate
Plate well missing from CSV

Suggested thresholds:

Ct > 35 = High Ct warning
Ct SD > 0.5 = Replicate variation warning
NTC Ct exists and Ct < 35 = NTC amplification warning

Thresholds should be stored in analysis run.

19. Graphs

MVP graph:

Fold Change Bar Graph

Mobile graph behavior:

Swipe between target genes
Tap bar to see value
Show/hide control sample
Show fold change or log2 fold change
Export PNG

Keep graph editing simple.

20. Notes and Observations

Since mobile is primary, notes become very important.

MVP notes:

Text note
Note type
Timestamp
Author

Note types:

General
SetupObservation
UploadNote
AnalysisNote
Issue
Conclusion

Future:

Photo attachment
Voice note
Barcode scan
Reagent photo
Gel image attachment

21. Export

21.1 CSV Export

Export analysis results.

Include:

Experiment Name
Project
Date
Researcher
Sample
Target Gene
Reference Gene
Mean Target Ct
SD Target Ct
Mean Reference Ct

```
SD Reference Ct
Delta Ct
Delta Delta Ct
Fold Change
Log2 Fold Change
Warnings
```

Protect CSV export from formula injection.

If cell starts with:

```
=
+
-
@
```

prefix it with apostrophe.

21.2 PDF Export

PDF should include:

```
Lab name
Project name
Experiment name
Researcher
Date
Objective
Reference gene
Control sample
Uploaded file name
Plate summary
Analysis table
Graph
Warnings
Excluded wells
Notes/conclusion
Generated date/time
```

21.3 PNG Export

Export graph image as PNG.

22. Database Design

Database: MS SQL Server

Use:

```
uniqueidentifier primary keys  
datetime2 timestamps  
decimal for Ct/calculation values  
nvarchar for strings  
bit for boolean flags  
soft delete for important records
```

22.1 Core Tables

```
Users  
Labs  
LabUsers  
Projects  
ProjectUsers  
Experiments  
ExperimentTargets  
PlateLayouts  
PlateWells  
UploadedFiles  
RawCtData  
AnalysisRuns  
AnalysisResults  
AnalysisWarnings  
ExperimentNotes  
ExportFiles  
AuditLogs
```

22.2 Users

```
CREATE TABLE Users (  
    UserId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,  
    Email NVARCHAR(255) NOT NULL UNIQUE,  
    PasswordHash NVARCHAR(MAX) NULL,  
    FullName NVARCHAR(200) NOT NULL,  
    SystemRole NVARCHAR(50) NOT NULL DEFAULT 'User',  
    IsActive BIT NOT NULL DEFAULT 1,  
    CreatedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),  
    UpdatedAt DATETIME2 NULL  
);
```

22.3 Labs

```
CREATE TABLE Labs (  
    LabId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,  
    LabName NVARCHAR(200) NOT NULL,  
    InstitutionName NVARCHAR(200) NULL,  
    Description NVARCHAR(MAX) NULL,  
    IsActive BIT NOT NULL DEFAULT 1,  
    CreatedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),  
    UpdatedAt DATETIME2 NULL  
);
```

22.4 LabUsers

```
CREATE TABLE LabUsers (  
    LabUserId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,  
    LabId UNIQUEIDENTIFIER NOT NULL,  
    UserId UNIQUEIDENTIFIER NOT NULL,  
    LabRole NVARCHAR(50) NOT NULL DEFAULT 'Researcher',  
    IsActive BIT NOT NULL DEFAULT 1,  
    JoinedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),  
  
    CONSTRAINT FK_LabUsers_Labs  
        FOREIGN KEY (LabId) REFERENCES Labs(LabId),  
  
    CONSTRAINT FK_LabUsers_Users  
        FOREIGN KEY (UserId) REFERENCES Users(UserId),
```

```
CONSTRAINT UQ_LabUsers_Lab_User
    UNIQUE (LabId, UserId)
);
```

LabRole values:

```
LabAdmin
PI
Researcher
Viewer
```

22.5 Projects

```
CREATE TABLE Projects (
    ProjectId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,
    LabId UNIQUEIDENTIFIER NOT NULL,
    ProjectName NVARCHAR(200) NOT NULL,
    Description NVARCHAR(MAX) NULL,
    CreatedByUserId UNIQUEIDENTIFIER NOT NULL,
    IsDeleted BIT NOT NULL DEFAULT 0,
    CreatedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),
    UpdatedAt DATETIME2 NULL,

    CONSTRAINT FK_Projects_Labs
        FOREIGN KEY (LabId) REFERENCES Labs(LabId),

    CONSTRAINT FK_Projects_Users
        FOREIGN KEY (CreatedByUserId) REFERENCES Users(UserId)
);
```

22.6 ProjectUsers

```
CREATE TABLE ProjectUsers (
    ProjectUserId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,
    ProjectId UNIQUEIDENTIFIER NOT NULL,
    UserId UNIQUEIDENTIFIER NOT NULL,
    ProjectRole NVARCHAR(50) NOT NULL DEFAULT 'Member',
    IsActive BIT NOT NULL DEFAULT 1,
    CreatedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),
```

```

CONSTRAINT FK_ProjectUsers_Projects
    FOREIGN KEY (ProjectId) REFERENCES Projects(ProjectId),

CONSTRAINT FK_ProjectUsers_Users
    FOREIGN KEY (UserId) REFERENCES Users(UserId),

CONSTRAINT UQ_ProjectUsers_Project_User
    UNIQUE (ProjectId, UserId)
);

```

22.7 Experiments

```

CREATE TABLE Experiments (
    ExperimentId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,
    LabId UNIQUEIDENTIFIER NOT NULL,
    ProjectId UNIQUEIDENTIFIER NULL,
    OwnerUserId UNIQUEIDENTIFIER NOT NULL,
    ExperimentName NVARCHAR(200) NOT NULL,
    ExperimentDate DATE NOT NULL,
    Objective NVARCHAR(MAX) NULL,
    Hypothesis NVARCHAR(MAX) NULL,
    SampleSource NVARCHAR(300) NULL,
    TreatmentCondition NVARCHAR(300) NULL,
    InstrumentName NVARCHAR(200) NULL,
    ReferenceGene NVARCHAR(100) NULL,
    ControlSampleName NVARCHAR(200) NULL,
    PlateType NVARCHAR(50) NOT NULL DEFAULT '96-well',
    Status NVARCHAR(50) NOT NULL DEFAULT 'Draft',
    Visibility NVARCHAR(50) NOT NULL DEFAULT 'Lab',
    Notes NVARCHAR(MAX) NULL,
    LockedAt DATETIME2 NULL,
    LockedByUserId UNIQUEIDENTIFIER NULL,
    IsArchived BIT NOT NULL DEFAULT 0,
    IsDeleted BIT NOT NULL DEFAULT 0,
    CreatedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),
    UpdatedAt DATETIME2 NULL,

    CONSTRAINT FK_Experiments_Labs
        FOREIGN KEY (LabId) REFERENCES Labs(LabId),

    CONSTRAINT FK_Experiments_Projects
        FOREIGN KEY (ProjectId) REFERENCES Projects(ProjectId),

```

```
CONSTRAINT FK_Experiments_Owner
    FOREIGN KEY (OwnerUserId) REFERENCES Users(UserId)
);
```

22.8 PlateLayouts

```
CREATE TABLE PlateLayouts (
    PlateLayoutId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,
    ExperimentId UNIQUEIDENTIFIER NOT NULL,
    PlateType NVARCHAR(50) NOT NULL DEFAULT '96-well',
    RowCount INT NOT NULL DEFAULT 8,
    ColumnCount INT NOT NULL DEFAULT 12,
    CreatedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),
    UpdatedAt DATETIME2 NULL,

    CONSTRAINT FK_PlateLayouts_Experiments
        FOREIGN KEY (ExperimentId) REFERENCES Experiments(ExperimentId)
);
```

22.9 PlateWells

```
CREATE TABLE PlateWells (
    PlateWellId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,
    PlateLayoutId UNIQUEIDENTIFIER NOT NULL,
    ExperimentId UNIQUEIDENTIFIER NOT NULL,
    WellId NVARCHAR(5) NOT NULL,
    RowLabel NVARCHAR(2) NOT NULL,
    ColumnNumber INT NOT NULL,
    SampleName NVARCHAR(200) NULL,
    TargetGene NVARCHAR(100) NULL,
    ReferenceGene NVARCHAR(100) NULL,
    SampleType NVARCHAR(50) NULL,
    ReplicateGroup NVARCHAR(100) NULL,
    CtValue DECIMAL(10,4) NULL,
    IsExcluded BIT NOT NULL DEFAULT 0,
    ExclusionReason NVARCHAR(300) NULL,
    Notes NVARCHAR(MAX) NULL,
    CreatedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),
    UpdatedAt DATETIME2 NULL,

    CONSTRAINT FK_PlateWells_PlateLayouts
```

```

        FOREIGN KEY (PlateLayoutId) REFERENCES PlateLayouts(PlateLayoutId),

CONSTRAINT FK_PlateWells_Experiments
        FOREIGN KEY (ExperimentId) REFERENCES Experiments(ExperimentId),

CONSTRAINT UQ_PlateWells_Layout_Well
        UNIQUE (PlateLayoutId, WellId)
);

```

22.10 UploadedFiles

```

CREATE TABLE UploadedFiles (
    UploadedFileId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,
    ExperimentId UNIQUEIDENTIFIER NOT NULL,
    OriginalFileName NVARCHAR(300) NOT NULL,
    StoredFilePath NVARCHAR(1000) NOT NULL,
    FileType NVARCHAR(50) NOT NULL,
    FileSizeBytes BIGINT NOT NULL,
    UploadedByUserId UNIQUEIDENTIFIER NOT NULL,
    UploadedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),

CONSTRAINT FK_UploadedFiles_Experiments
        FOREIGN KEY (ExperimentId) REFERENCES Experiments(ExperimentId),

CONSTRAINT FK_UploadedFiles_Users
        FOREIGN KEY (UploadedByUserId) REFERENCES Users(UserId)
);

```

22.11 RawCtData

```

CREATE TABLE RawCtData (
    RawCtDataId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,
    ExperimentId UNIQUEIDENTIFIER NOT NULL,
    UploadedFileId UNIQUEIDENTIFIER NOT NULL,
    WellId NVARCHAR(5) NOT NULL,
    SampleNameFromFile NVARCHAR(200) NULL,
    TargetGeneFromFile NVARCHAR(100) NULL,
    CtValueRaw NVARCHAR(100) NULL,
    CtValue DECIMAL(10,4) NULL,
    IsValid BIT NOT NULL DEFAULT 1,
    ValidationMessage NVARCHAR(500) NULL,

```

```

    CreatedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),

    CONSTRAINT FK_RawCtData_Experiments
        FOREIGN KEY (ExperimentId) REFERENCES Experiments(ExperimentId),

    CONSTRAINT FK_RawCtData_UploadedFiles
        FOREIGN KEY (UploadedFileId) REFERENCES UploadedFiles(UploadedFileId)
);

```

22.12 AnalysisRuns

```

CREATE TABLE AnalysisRuns (
    AnalysisRunId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,
    ExperimentId UNIQUEIDENTIFIER NOT NULL,
    RunName NVARCHAR(200) NULL,
    ControlSampleName NVARCHAR(200) NOT NULL,
    ReferenceGene NVARCHAR(100) NOT NULL,
    CalculationMethod NVARCHAR(100) NOT NULL DEFAULT 'DeltaDeltaCt',
    HighCtThreshold DECIMAL(10,4) NOT NULL DEFAULT 35,
    ReplicateSdThreshold DECIMAL(10,4) NOT NULL DEFAULT 0.5,
    CreatedByUserId UNIQUEIDENTIFIER NOT NULL,
    CreatedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),

    CONSTRAINT FK_AnalysisRuns_Experiments
        FOREIGN KEY (ExperimentId) REFERENCES Experiments(ExperimentId),

    CONSTRAINT FK_AnalysisRuns_Users
        FOREIGN KEY (CreatedByUserId) REFERENCES Users(UserId)
);

```

22.13 AnalysisResults

```

CREATE TABLE AnalysisResults (
    AnalysisResultId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,
    AnalysisRunId UNIQUEIDENTIFIER NOT NULL,
    ExperimentId UNIQUEIDENTIFIER NOT NULL,
    SampleName NVARCHAR(200) NOT NULL,
    TargetGene NVARCHAR(100) NOT NULL,
    ReferenceGene NVARCHAR(100) NOT NULL,
    SampleType NVARCHAR(50) NULL,
    MeanTargetCt DECIMAL(10,4) NULL,

```

```

SdTargetCt DECIMAL(10,4) NULL,
MeanReferenceCt DECIMAL(10,4) NULL,
SdReferenceCt DECIMAL(10,4) NULL,
DeltaCt DECIMAL(10,4) NULL,
DeltaDeltaCt DECIMAL(10,4) NULL,
FoldChange DECIMAL(18,8) NULL,
Log2FoldChange DECIMAL(18,8) NULL,
TargetReplicateCount INT NOT NULL DEFAULT 0,
ReferenceReplicateCount INT NOT NULL DEFAULT 0,
HasWarning BIT NOT NULL DEFAULT 0,
CreatedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),

CONSTRAINT FK_AnalysisResults_AnalysisRuns
    FOREIGN KEY (AnalysisRunId) REFERENCES AnalysisRuns(AnalysisRunId),

CONSTRAINT FK_AnalysisResults_Experiments
    FOREIGN KEY (ExperimentId) REFERENCES Experiments(ExperimentId)
);

```

22.14 AnalysisWarnings

```

CREATE TABLE AnalysisWarnings (
    AnalysisWarningId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,
    AnalysisRunId UNIQUEIDENTIFIER NOT NULL,
    AnalysisResultId UNIQUEIDENTIFIER NULL,
    PlateWellId UNIQUEIDENTIFIER NULL,
    WarningType NVARCHAR(100) NOT NULL,
    WarningMessage NVARCHAR(500) NOT NULL,
    Severity NVARCHAR(50) NOT NULL DEFAULT 'Warning',
    CreatedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),

CONSTRAINT FK_AnalysisWarnings_AnalysisRuns
    FOREIGN KEY (AnalysisRunId) REFERENCES AnalysisRuns(AnalysisRunId),

CONSTRAINT FK_AnalysisWarnings_AnalysisResults
    FOREIGN KEY (AnalysisResultId) REFERENCES
AnalysisResults(AnalysisResultId),

CONSTRAINT FK_AnalysisWarnings_PlateWells
    FOREIGN KEY (PlateWellId) REFERENCES PlateWells(PlateWellId)
);

```

22.15 ExperimentNotes

```
CREATE TABLE ExperimentNotes (  
    ExperimentNoteId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,  
    ExperimentId UNIQUEIDENTIFIER NOT NULL,  
    UserId UNIQUEIDENTIFIER NOT NULL,  
    NoteText NVARCHAR(MAX) NOT NULL,  
    NoteType NVARCHAR(50) NOT NULL DEFAULT 'General',  
    CreatedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),  
  
    CONSTRAINT FK_ExperimentNotes_Experiments  
        FOREIGN KEY (ExperimentId) REFERENCES Experiments(ExperimentId),  
  
    CONSTRAINT FK_ExperimentNotes_Users  
        FOREIGN KEY (UserId) REFERENCES Users(UserId)  
);
```

22.16 ExportFiles

```
CREATE TABLE ExportFiles (  
    ExportFileId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,  
    ExperimentId UNIQUEIDENTIFIER NOT NULL,  
    AnalysisRunId UNIQUEIDENTIFIER NULL,  
    ExportType NVARCHAR(50) NOT NULL,  
    FileName NVARCHAR(300) NOT NULL,  
    FilePath NVARCHAR(1000) NOT NULL,  
    CreatedByUserId UNIQUEIDENTIFIER NOT NULL,  
    CreatedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME(),  
  
    CONSTRAINT FK_ExportFiles_Experiments  
        FOREIGN KEY (ExperimentId) REFERENCES Experiments(ExperimentId),  
  
    CONSTRAINT FK_ExportFiles_AnalysisRuns  
        FOREIGN KEY (AnalysisRunId) REFERENCES AnalysisRuns(AnalysisRunId),  
  
    CONSTRAINT FK_ExportFiles_Users  
        FOREIGN KEY (CreatedByUserId) REFERENCES Users(UserId)  
);
```

22.17 AuditLogs

```
CREATE TABLE AuditLogs (  
  AuditLogId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,  
  LabId UNIQUEIDENTIFIER NULL,  
  UserId UNIQUEIDENTIFIER NULL,  
  EntityName NVARCHAR(100) NOT NULL,  
  EntityId UNIQUEIDENTIFIER NULL,  
  Action NVARCHAR(100) NOT NULL,  
  OldValue NVARCHAR(MAX) NULL,  
  NewValue NVARCHAR(MAX) NULL,  
  CreatedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME()  
);
```

23. API Design

Base path:

```
/api
```

23.1 Authentication APIs

```
POST /api/auth/login  
POST /api/auth/register  
GET /api/auth/me
```

23.2 Dashboard APIs

```
GET /api/dashboard/mobile  
GET /api/dashboard/my  
GET /api/dashboard/lab  
GET /api/dashboard/project/{projectId}
```

23.3 Lab APIs

```
GET    /api/labs/current
GET    /api/labs/{labId}/users
POST   /api/labs/{labId}/users/invite
PUT    /api/labs/{labId}/users/{userId}/role
```

23.4 Project APIs

```
GET     /api/projects
POST    /api/projects
GET     /api/projects/{projectId}
PUT     /api/projects/{projectId}
DELETE  /api/projects/{projectId}
GET     /api/projects/{projectId}/experiments
```

23.5 Experiment APIs

```
GET     /api/experiments
POST    /api/experiments
GET     /api/experiments/{experimentId}
PUT     /api/experiments/{experimentId}
DELETE  /api/experiments/{experimentId}
POST    /api/experiments/{experimentId}/duplicate
POST    /api/experiments/{experimentId}/finalize
POST    /api/experiments/{experimentId}/unlock
```

23.6 Plate APIs

```
GET     /api/experiments/{experimentId}/plate
GET     /api/experiments/{experimentId}/plate/list
PUT     /api/experiments/{experimentId}/plate/wells/{plateWellId}
PUT     /api/experiments/{experimentId}/plate/wells/bulk
POST    /api/experiments/{experimentId}/plate/quick-fill
POST    /api/experiments/{experimentId}/plate/upload-layout
POST    /api/experiments/{experimentId}/plate/wells/clear
```

```
POST /api/experiments/{experimentId}/plate/wells/exclude
POST /api/experiments/{experimentId}/plate/wells/include
```

23.7 Upload APIs

```
POST /api/experiments/{experimentId}/uploads/csv
POST /api/experiments/{experimentId}/uploads/{uploadedFileId}/map
GET /api/experiments/{experimentId}/uploads
```

23.8 Analysis APIs

```
POST /api/experiments/{experimentId}/analysis/run
GET /api/experiments/{experimentId}/analysis/latest
GET /api/experiments/{experimentId}/analysis/latest/mobile-summary
GET /api/analysis-runs/{analysisRunId}
GET /api/analysis-runs/{analysisRunId}/results
GET /api/analysis-runs/{analysisRunId}/warnings
```

23.9 Notes APIs

```
GET /api/experiments/{experimentId}/notes
POST /api/experiments/{experimentId}/notes
```

23.10 Export APIs

```
GET /api/analysis-runs/{analysisRunId}/export/csv
POST /api/analysis-runs/{analysisRunId}/export/pdf
POST /api/analysis-runs/{analysisRunId}/export/png
GET /api/exports/{exportFileId}/download
```

24. Important API Examples

24.1 Quick Fill Request

```
{
  "fromWell": "A01",
  "toWell": "A03",
  "sampleName": "S1",
  "targetGene": "CCR2",
  "referenceGene": "GAPDH",
  "sampleType": "Test",
  "replicateGroup": "S1_CCR2"
}
```

Rules:

```
Validate well range.
Update all wells in range.
Do not allow update if experiment is finalized.
Log audit entry.
```

24.2 Mobile Dashboard Response

```
{
  "summary": {
    "myDrafts": 3,
    "pendingAnalysis": 5,
    "experimentsWithWarnings": 2,
    "recentReports": 4
  },
  "recentExperiments": [
    {
      "experimentId": "guid",
      "experimentName": "CCR2_Repeat_1",
      "projectName": "CCR2 Study",
      "ownerName": "Kishi",
      "status": "Analyzed",
      "warningCount": 1,
      "lastUpdated": "2026-06-06T10:30:00Z"
    }
  ]
}
```

```
]
}
```

24.3 Mobile Analysis Summary Response

```
{
  "experimentName": "CCR2_Repeat_1",
  "referenceGene": "GAPDH",
  "controlSampleName": "Control",
  "targetGene": "CCR2",
  "warningCount": 2,
  "results": [
    {
      "sampleName": "S1",
      "deltaCt": 2.6,
      "deltaDeltaCt": 0.3,
      "foldChange": 0.81,
      "hasWarning": false
    },
    {
      "sampleName": "S2",
      "deltaCt": 3.3,
      "deltaDeltaCt": 1.0,
      "foldChange": 0.5,
      "hasWarning": true
    }
  ]
}
```

25. Backend Services

Create these services:

```
AuthService
LabService
ProjectService
ExperimentService
PermissionService
PlateService
CsvParsingService
CtMappingService
```

```
QpcrAnalysisService
GraphService
ExportService
ExperimentNoteService
AuditLogService
FileStorageService
```

26. Helper Classes

26.1 WellIdHelper

Methods:

```
NormalizeWellId(input)
IsValidWellId(input)
Generate96WellIds()
GenerateWellRange(fromWell, toWell)
```

Examples:

```
A1   -> A01
A01  -> A01
a1   -> A01
H12  -> H12
```

Invalid:

```
I01
A13
AA1
1A
```

26.2 MathHelper

Methods:

```
Mean(values)
SampleStandardDeviation(values)
FoldChange(deltaDeltaCt)
```

Fold change:

```
FoldChange = Math.Pow(2, -deltaDeltaCt)
```

27. Development Phases for Copilot

Phase 1: Backend Foundation

Tasks:

1. Create ASP.NET Core 8 Web API.
2. Configure MS SQL.
3. Add EF Core.
4. Create entities:
5. Users
6. Labs
7. LabUsers
8. Projects
9. ProjectUsers
10. Experiments
11. PlateLayouts
12. PlateWells
13. Add migrations.
14. Seed one lab, one admin, one researcher.
15. Add JWT authentication.

Acceptance criteria:

```
Backend runs.
Database is created.
User can log in.
User belongs to a lab.
```

Phase 2: Mobile Frontend Shell

Tasks:

1. Create Ionic React TypeScript project.
2. Add login page.
3. Add bottom navigation.
4. Add mobile lab dashboard page.
5. Add projects page.
6. Add experiments page.

Acceptance criteria:

Mobile UI loads.
User can log in.
Bottom tabs work.
Dashboard shows summary cards.

Phase 3: Experiment Creation

Tasks:

1. Create project APIs.
2. Create experiment APIs.
3. Build mobile create experiment screen.
4. Auto-create 96 wells when experiment is created.
5. Show experiment in My Experiments and Lab Dashboard.

Acceptance criteria:

User can create experiment from phone.
96 wells are created automatically.
Experiment appears in dashboards.

Phase 4: Mobile Plate Setup

Tasks:

1. Create plate list API.
2. Create quick-fill API.
3. Create plate layout CSV upload API.

4. Build Plate Setup Method screen.
5. Build Quick Fill screen.
6. Build Plate List View.
7. Build Plate Review Grid.
8. Build Manual Well Entry screen.

Acceptance criteria:

User can set up plate without desktop.
User can quick-fill well ranges.
User can upload plate layout CSV.
User can review plate as list and grid.

Phase 5: Ct CSV Upload

Tasks:

1. Add Ct CSV upload API.
2. Save uploaded file.
3. Parse CSV.
4. Return mobile-friendly preview.
5. Build column mapping screen.
6. Map Ct values to wells.
7. Generate validation warnings.

Acceptance criteria:

User can upload Ct CSV from mobile.
User can map Well and Ct columns.
Ct values are assigned correctly.
Warnings are displayed.

Phase 6: qPCR Analysis Engine

Tasks:

1. Build QpcrAnalysisService.
2. Implement mean Ct.
3. Implement standard deviation.
4. Implement ΔCt .
5. Implement $\Delta\Delta Ct$.
6. Implement fold change.

7. Implement log2 fold change.
8. Generate warnings.
9. Save analysis run.
10. Build mobile analysis summary screen.

Acceptance criteria:

User can run analysis.
Results match Excel.
Control fold change = 1.
Excluded wells are ignored.
Warnings are shown.

Phase 7: Graphs, Export, Notes

Tasks:

1. Build mobile graph screen.
2. Add CSV export.
3. Add PNG export.
4. Add PDF export.
5. Add notes APIs.
6. Build mobile notes screen.
7. Add audit logs.

Acceptance criteria:

User can view graph on mobile.
User can export results.
User can add notes.
PDF includes metadata, table, graph, warnings, and notes.

Phase 8: Finalization and Testing

Tasks:

1. Add finalize API.
2. Add unlock API.
3. Block edits after finalization.
4. Test permission rules.
5. Test with real qPCR files from Kishi.
6. Compare app results with Excel.

Acceptance criteria:

Finalized experiment is read-only.
PI/Admin can unlock.
Real experiment matches Excel results.
Mobile workflow is usable end to end.

28. MVP Success Criteria

The MVP is successful if:

A researcher can use mobile phone to create an experiment.
A researcher can set up plate layout without desktop.
A researcher can upload Ct CSV.
The app maps Ct values correctly.
The app calculates ΔCt , $\Delta\Delta Ct$, and fold change correctly.
Results match existing Excel calculations.
Graphs are generated automatically.
Reports can be exported.
Other lab members can see shared experiments.
Lab dashboard gives useful visibility.
Notes preserve observations.
Finalized experiments are protected.

29. What Not to Build in MVP

Do not build in MVP:

Real-time collaboration
Full electronic lab notebook
Inventory management
Primer database
AI analysis
Direct qPCR machine integration
Advanced statistics
Regulatory compliance workflow
External collaborator portal
PowerPoint export

384-well support
Native mobile app

Ionic PWA is enough first.

30. First Copilot Prompt

Use this as the first Copilot task:

Create an ASP.NET Core 8 Web API project using Entity Framework Core and MS SQL Server.

Create entities and DbContext for Users, Labs, LabUsers, Projects, ProjectUsers, Experiments, PlateLayouts, and PlateWells.

When a new experiment is created, automatically create one 96-well PlateLayout and 96 PlateWells from A01 to H12.

Add JWT authentication and basic lab-level permission checks.

Do not build frontend yet.

Second Copilot task:

Create an Ionic React TypeScript mobile-first frontend with login page, bottom navigation, mobile lab dashboard, projects page, and experiments page.

Use card-based layouts, not desktop tables.

31. Final Recommendation

Build the product as:

Mobile-first shared qPCR lab workflow app

The most important rules are:

Use LabId from day one.
Use Visibility from day one.
Use Finalized/Locked status from day one.
Do not make desktop grid editing the main workflow.
Use Quick Fill, Plate Layout CSV upload, Copy Previous Experiment, and Plate List View for mobile.

This design will support real lab usage much better than a desktop-only qPCR calculator.